

Policy Strictness and Heterogeneous Mobility Responses: A Natural Experiment on Workplace and Park Mobility in Denmark and Sweden

1. Introduction

1.1 Research Background

During the initial outbreak of COVID-19 in early 2020, Denmark and Sweden, two Nordic welfare states with highly similar socio-economic structures, adopted strikingly different policy responses. Denmark implemented a strict, mandatory legal lockdown starting March 13, 2020, which included school closures and restrictions on public gatherings. In contrast, Sweden chose a unique path of voluntary recommendations and herd immunity strategies, keeping most schools and businesses open. This policy divergence presents a unique “natural experiment”, allowing us to isolate the causal effect of policy strictness (Mandatory vs. Voluntary) on human behavior.

1.2 Research Objectives

Current research frequently analyzes the epidemiological outcomes of lockdowns and their aggregate impact on human mobility. However, less attention has been paid to the uneven impact of these policies across different regions. Crucial variables, such as employment rates and poverty rates, likely dictate whether individuals prioritize economic necessity (work) or psychological relief (leisure). This study fills this gap by moving beyond average effects to analyze how specific regional characteristics drive divergent mobility behaviors.

- RQ1: How did the mandatory lockdown in Denmark, compared to Sweden’s voluntary strategy, affect workplace and park mobility?
- RQ2: How do regional characteristics (employment rate, poverty rate) shape heterogeneous mobility responses to the lockdown?

1.3 Hypotheses

This study proposes two main sets of hypotheses regarding workplace and park mobility:

- **H1 (Workplace Mobility):** The mandatory lockdown in Denmark results in a significantly larger decline in workplace mobility compared to the voluntary recommendations in Sweden.

Explanation: Multiple studies provide evidence that Denmark's mandatory lockdown resulted in a significantly larger decline in workplace mobility compared to Sweden's reliance on voluntary recommendations. Zhang et al. (2021) analyzed Google Community Mobility Reports. They found that Denmark experienced a sharper initial reduction in workplace mobility following the implementation of its legal lockdown, whereas Sweden's decline in mobility was less pronounced, despite voluntary recommendations. Similarly, Mishra et al. (2021) compared mobility changes across several countries and concluded that lockdowns led to greater reductions in workplace mobility than voluntary measures, with Denmark's response causing a more immediate and substantial drop than Sweden's. Born et al. (2021) further demonstrated, through synthetic control analysis, that actual mobility reductions in Sweden, although notable, were less severe than those observed in Denmark under a lockdown scenario, highlighting the more substantial effect of mandatory policies on workplace mobility.

- **H1a (Heterogeneity):** The lockdown's impact is heterogeneous. Specifically, regions with **higher employment rates** are expected to experience a sharper decline in workplace mobility, as the policy shock disproportionately affects economic hubs.

Explanation: Our analysis builds upon previous research that highlights regional disparities in employment and mobility in Denmark. Nordregio's "State of the Nordic Region 2024" reports that employment rates are highest in urban hubs, such as the Capital Region (Hovedstaden), while peripheral regions, like Region Sjælland and Region Nordjylland, have lower

employment rates. The report further indicates that urban areas experienced sharper declines in workplace mobility during the pandemic, consistent with our hypothesis. Similarly, the OECD's "Mobility and Integrated Labour Markets for Third-country Nationals in Greater Copenhagen" notes that Greater Copenhagen (Capital Region) has the highest employment rate in Denmark and experienced significant mobility changes during the pandemic, emphasizing that policy shocks disproportionately affect regions with higher employment and economic activity.

- **H2 (Park Mobility):** During the lockdown period, park mobility increases in Denmark relative to Sweden, because parks function as a "grey zone" that enables relatively safer outdoor activity when indoor options are restricted.

Explanation: Zhang et al. (2021) relied on aggregated mobility data, which often obscured the distinction between the cessation of indoor activities and the increase in outdoor leisure activities. By specifically isolating park mobility, this study addresses these limitations and more directly confirms the critical role of parks as relatively safe "grey zones" during the lockdown, rather than treating them merely as a component of overall reduced mobility. This approach yields more precise policy insights, helping governments better balance infection control with citizens' needs for essential outdoor resources.

- **H2a (Heterogeneity):** During the lockdown period, high-poverty municipalities exhibit a smaller increase (or a larger decline) in park mobility compared to low-poverty municipalities, indicating unequal access to and use of urban green spaces.

Explanation: This hypothesis examines how socioeconomic disadvantage shapes the fairness of park use during the pandemic, drawing on the multidimensional analysis in Geng et al. (2021). They show that park visitation tends to be higher in more economically developed contexts and argue that weaker economic performance can restrict access to parks and intensify the negative impacts of COVID-19. In line with this, the U.S.-based studies they reference find that access to urban green spaces is negatively

associated with poverty rates (Larson et al., 2016; Laster Pirtle, 2020; Nesbitt et al., 2019). Building on this body of evidence, H2a posits that areas with higher poverty rates will exhibit more pronounced suppression of growth in park mobility during the pandemic, reflecting structural inequalities in access to parks and other urban green spaces.

2. Research Design

This study uses a Difference-in-Differences (DiD) design to estimate the impact of Denmark's emergency lockdown on mobility, using Sweden as the comparison country. The analysis addresses two questions: RQ1 (average lockdown effect on workplace and park mobility) and RQ2 (whether the effect varies by regional socioeconomic conditions).

2.1 Identification Strategy

Treatment group: municipalities in Denmark (legal lockdown). Control group: municipalities in Sweden (voluntary recommendations). The policy window is defined as 2020/03/13–2020/04/20 (lockdown), surrounded by non-lockdown periods (2020/02/13–2020/03/12 and 2020/04/21–2020/05/13). Under the parallel trends assumption, absent the lockdown, the Denmark–Sweden mobility gap would have evolved similarly across the lockdown window and the surrounding non-lockdown windows; therefore, the change in the Denmark–Sweden gap from 2020/03/13 to 2020/04/20 identifies the causal lockdown effect.

2.2 Treatment and Outcome Variables

The outcome Y_{it} is the daily percentage change in mobility from baseline for municipality i on day t , measured separately for workplace mobility and park mobility. We define:

- $Treat_i = 1$ for Danish regions and 0 for Swedish regions.
- $Post_t = 1$ if day t falls in 2020/03/13–2020/04/20, and 0 if in the non-lockdown periods.
- $D_{it} = Treat_i \times Post_t$

To control for shocks standard to both countries on each day, we include date fixed effects λ_t ; therefore, the standalone $Post_t$ is absorbed and is used only to construct D_{it}

2.3 Heterogeneity Variables

To test whether the Denmark–Sweden lockdown effect differs across regional conditions, we interact the DiD term D_{it} with a binary group indicator G_i (high vs. low). G_i is constructed within each country using that country’s median:

- $G_i^{Emp} = 1$ for high-employment regions and 0 for low-employment regions.
- $G_i^{Pov} = 1$ for high-poverty regions and 0 for low-poverty regions.

3. Data

3.1 Data Sources

- **Mobility:** Google Community Mobility Reports (municipality-level daily workplace and park mobility).
- **Policy timing/stringency check:** Oxford COVID-19 Government Response Tracker (OxCGRT).
- **Regional characteristics (moderators):** employment rate and poverty rate from official national statistics (Denmark and Sweden), measured under the same EU standards.

3.2 Sample Construction

We use daily data from 2020/02/13–2020/05/13, covering the pre-lockdown window, the lockdown window (2020/03/13–2020/04/20), and the short-run post-lockdown window (2020/04/21–2020/05/13). Mobility values suppressed by Google due to very low traffic or insufficient data are imputed as 0, treating these cases as no observable change in mobility for that region-day.

3.3 Key Variables

- Outcomes: workplace mobility; park mobility (% change from baseline).
- Treatment: $Treat_{it}$, $Post_{it}$, and $D_{it} = Treat_{it} \times Post_{it}$
- Moderators: employment rate, poverty rate, and binary group indicators G_i^{Emp} , G_i^{Pov} based on within-country medians.

4. Methodology

4.1 Baseline Difference-in-Differences (H1/H2)

We estimate the average Denmark–Sweden lockdown effect using a DiD specification with date fixed effects. For each outcome, Y_{it} (workplace mobility or park mobility), we estimate:

$$Y_{it} = \alpha + \tau D_{it} + \beta_1 Treat_i + \lambda_t + \varepsilon_{it}$$

Where the DiD term is defined as:

$$D_{it} = Treat_i \times Post_t$$

Because we include date fixed effects λ_t , the main effect of $Post_t$ is absorbed and is therefore not included as a separate regressor; $Post_t$ is used only to construct D_{it} . The coefficient τ captures how the Denmark–Sweden mobility gap changes during the lockdown window relative to the surrounding non-lockdown periods. Standard errors are clustered at the municipality level.

4.2 Interaction DiD Heterogeneity Model (H1a/H2a)

To test whether the Denmark–Sweden lockdown effect varies across regional socioeconomic conditions, we interact the DiD term with a binary group indicator G_i (high vs. low). For each outcome Y_{it} , we estimate:

$$Y_{it} = \alpha + \tau D_{it} + \beta_1 Treat_i + \beta_2 G_i + \delta(D_{it} \times G_i) + \lambda_t + \varepsilon_{it}$$

Here, τ is the Denmark–Sweden lockdown effect for the low group ($G_i = 0$). The interaction coefficient δ captures how much the lockdown effect differs for the high group ($G_i = 1$); therefore, the implied high-group effect is $\tau + \delta$. The term $\beta_2 G_i$ absorbs time-invariant level differences between high- and low-group municipalities, ensuring that δ is interpreted as heterogeneity in the lockdown effect rather than a baseline gap. Standard errors are clustered at the municipality level.

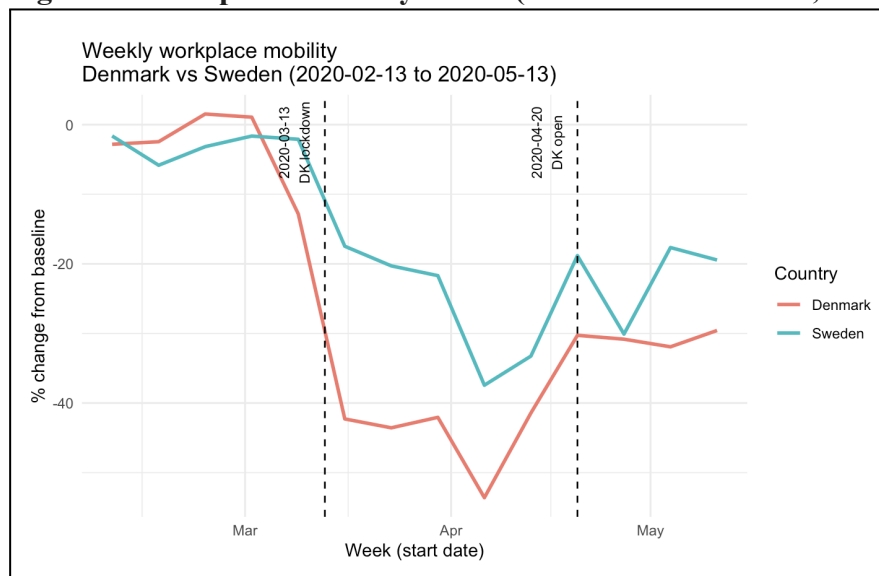
5. Results

5.1 Descriptive Analysis: Mobility Trends

Before presenting the regression estimates, we first visualize the raw trends in mobility patterns for Denmark (Treatment) and Sweden (Control) around the date of lockdown implementation.

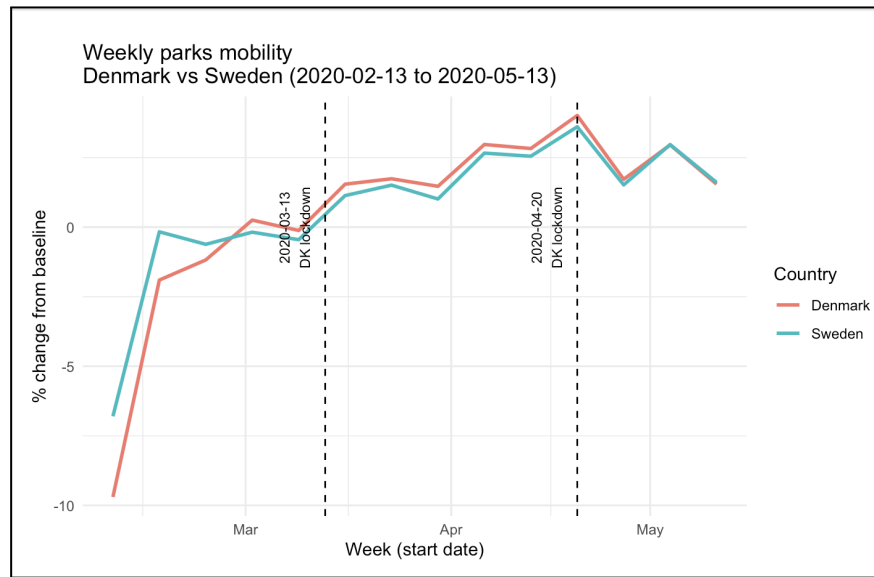
- **Figure 1** illustrates changes in workplace mobility around March 13, 2020, when Denmark formally implemented a legal lockdown in response to the global COVID-19 outbreak. Following the introduction of Denmark's strict lockdown measures, workplace mobility dropped sharply, with a much larger decline than in neighboring Sweden, which relied on less stringent, recommendation-based policies. After Denmark began to relax its restrictions on April 20, we observed a gradual increase in workplace mobility in Denmark, compared to Sweden.

Figure 1: Workplace Mobility Trend (Denmark vs. Sweden, 2/13-5/13)



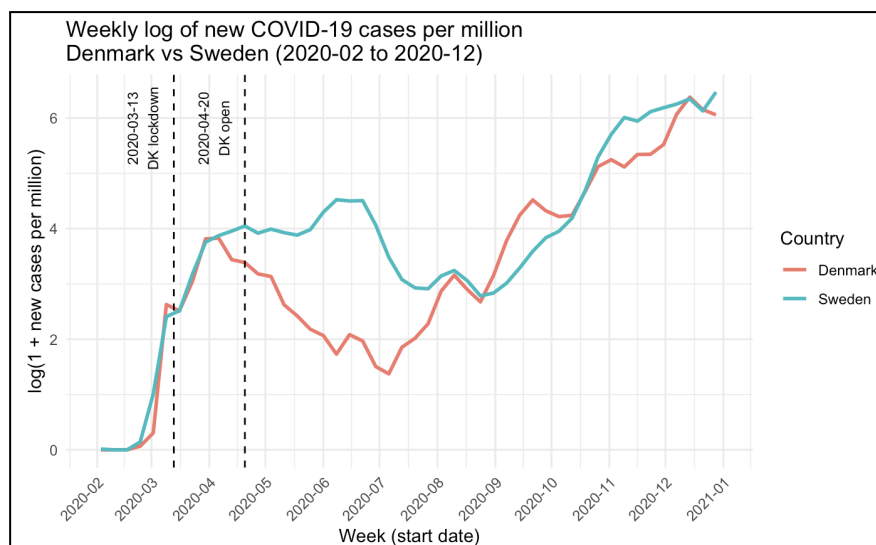
- **Figure 2** shows that, after the March 13 lockdown, both Denmark, with strict legal restrictions, and Sweden, with more lenient, recommendation-based measures, experienced a noticeable increase in visits to parks. In both countries, park mobility rose to roughly 5% above pre-pandemic baseline levels. This pattern suggests that, regardless of whether containment policies are strict or relatively relaxed, people still tend to seek outdoor green spaces such as parks to satisfy their need for fresh air and physical activity.

Figure 2: Park Mobility Trend (Denmark vs. Sweden, 2/13-5/13)



- Figure 3** shows that as the global situation escalated and COVID-19 was declared a pandemic on 11 March, the number of new confirmed cases in both Denmark and Sweden rose sharply. After Denmark formally implemented lockdown measures on 13 March, the upward trend in new cases began to flatten roughly two to three weeks later, and the subsequent decline was noticeably faster than in Sweden, which did not adopt similarly strict lockdown policies. However, in late July, the number of newly confirmed cases surged sharply.

Figure 3: New COVID-19 cases per million(2/13-12/31)



5.2 Baseline Estimates

This section reports baseline DiD estimates with date fixed effects. The coefficient τ on D_{it} captures the Denmark–Sweden lockdown effect for RQ1.

- **H1 (Workplace Mobility)**

H1 predicts that Denmark’s mandatory lockdown resulted in a larger decline in workplace mobility compared to Sweden’s voluntary approach. In Table 1, the DiD term D_{it} (Denmark $\times$ Lockdown) is large and negative ($\tau = -16.770$), indicating that workplace mobility in Denmark decreased by approximately 16.8 percentage points more than in Sweden during the lockdown window, after accounting for date-fixed effects. This effect is highly statistically significant ($p < 0.001$), and the 95% CI is entirely below zero ($[-17.672, -15.869]$), which provides strong evidence that the lockdown led to a substantially larger workplace decline in Denmark. Therefore, **H1 is supported.**

H2 (Park Mobility):

H2 predicts that during lockdown, Denmark’s park mobility should increase relative to Sweden (parks as a “grey zone” for safer outdoor activity). In Table 1, the estimated DiD effect is positive but small ($\tau = 0.361$). However, the evidence does not support a meaningful increase: the p-value is 0.572, and the 95% CI includes zero ($[-0.894, 1.616]$). This indicates the data do not provide strong statistical evidence that Denmark’s park mobility changed differently from Sweden’s during the lockdown window. Therefore, **H2 is not supported in the baseline DiD results.**

Table 1. Difference-in-Differences Regressions

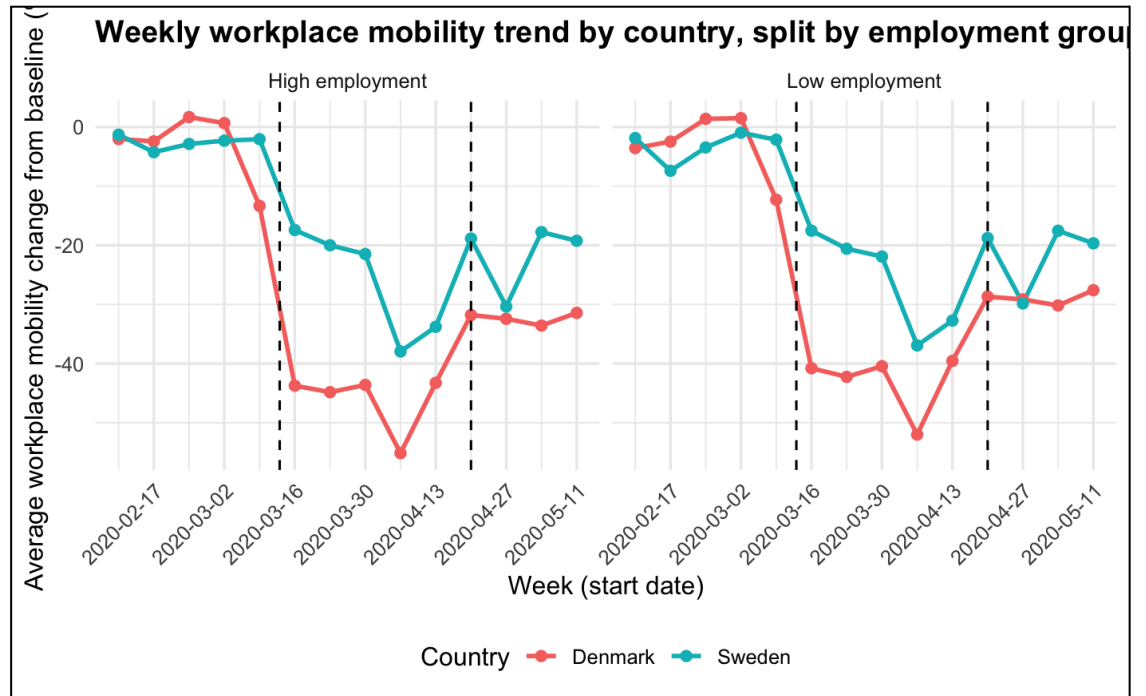
	Dependent variable	
	Workplace mobility	Park mobility
Intercept	-0.124 [-1.443, 1.196]	-0.246 [-2.072, 1.580]
	-0.184	-0.264
D_{it} (Denmark $\times$ Lockdown)	-16.770*** [-17.275, -16.266]	0.361 [-0.338, 1.059]
	-65.160	1.012
$Treat_i$ (Denmark)	-4.113*** [-4.447, -3.778]	-0.291 [-0.754, 0.172]
	-24.099	-1.233
Number of observations	25290	25290
R^2	0.806	0.048
Adj. R^2	0.805	0.044

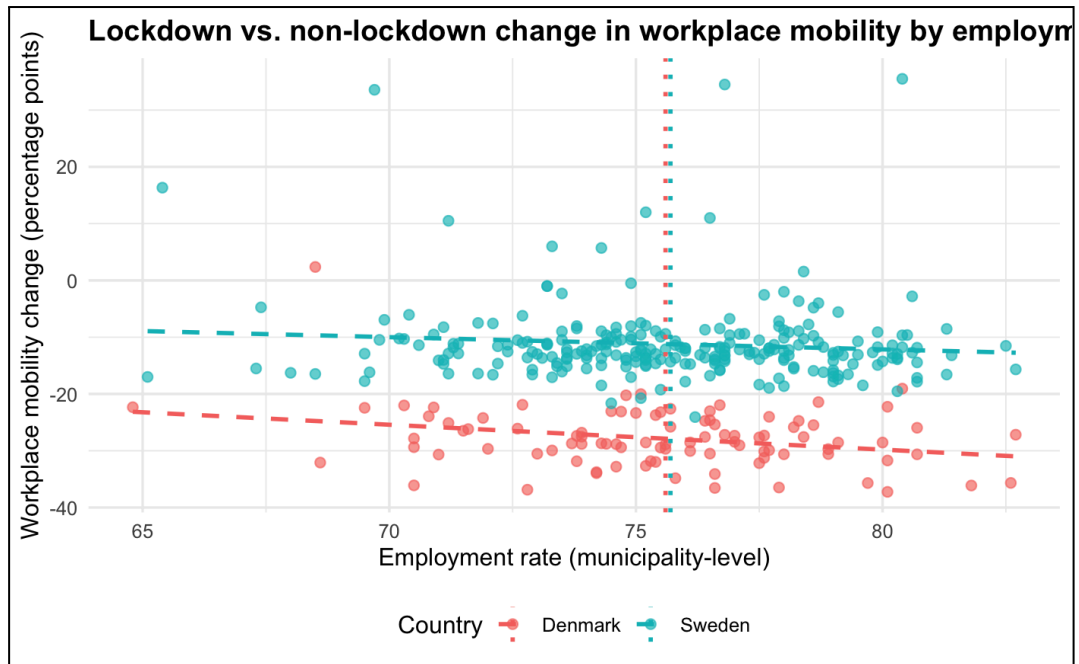
Note:

Cluster-robust standard errors (municipality level) are in parentheses. Two-sided p-values are shown next to SE and computed using a t distribution with $df = (\# \text{ clusters} - 1)$. 95% confidence intervals are in brackets. Date fixed effects are included but not reported. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

5.3 Interaction DiD Heterogeneity Model (H1a/H2a) estimates whether the Denmark–Sweden lockdown effect varies across municipalities by interacting the DiD term with within-country high vs. low employment (H1a) and high vs. low poverty (H2a) group indicators.

- H1a (Heterogeneity & Prediction):** H1a predicts that workplace decline will be more severe in high-employment regions. In Table 2 (workplace model), the baseline lockdown effect for the reference group ($G_i = 0$; low-employment regions) is strongly negative ($\tau = -15.413^*$, 95% CI $[-16.021, -14.805]$, implying $p < 0.01$). Importantly, the interaction term $D_{it} \times \text{High employment}$ is also negative and statistically significant ($\delta = -2.676^*$, 95% CI $[-3.347, -2.006]$, $p < 0.01$), indicating that the workplace decline in Denmark (relative to Sweden) becomes even more pronounced in regions with high employment. Substantively, this implies that the estimated lockdown effect in high-employment regions is approximately $-15.413 + (-2.676) = -18.089$, i.e., about 2.7% more decline than in low-employment regions. This aligns with the idea that the shock hits economic hubs harder. Therefore, **H1a is supported.**





H2a: Poverty and park mobility – observed pattern (opposite sign to prediction): H2a predicts that high-poverty municipalities should show a smaller increase (or larger decline) in park mobility (unequal access/usage of green space). Table 2 (park model) shows a statistically significant heterogeneity pattern, but in the opposite direction of H2a.

- For the reference group ($G_i = 0$; low-poverty municipalities), the lockdown effect is negative ($\tau = -0.928^{**}$, 95% CI $[-1.769, -0.087]$, implying $p < 0.05$), suggesting Denmark's park mobility declined relative to Sweden in low-poverty areas during lockdown.
- The interaction term $D_{it} \times \text{High poverty}$ is positive and highly significant ($\delta = 2.528^*$, 95% CI $[1.602, 3.454]$, $p < 0.01$), meaning that high-poverty municipalities experience a larger (more positive) lockdown effect than low-poverty ones.

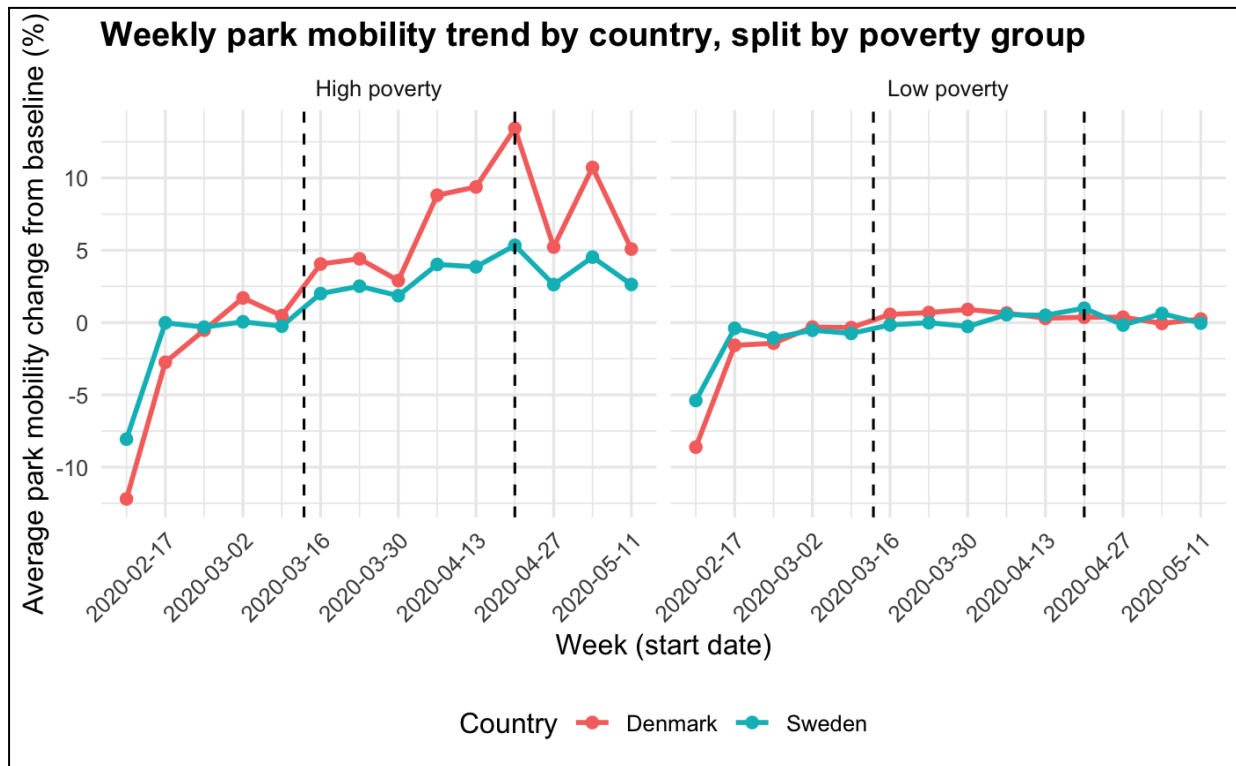
Combining the two terms implies the lockdown effect in high-poverty municipalities is approximately $-0.928 + 2.528 = +1.600$, i.e., a net increase relative to Sweden, contrary to the hypothesis of a smaller increase. Therefore, **H2a is not supported** (the heterogeneity exists, but the sign contradicts the expected inequality direction).

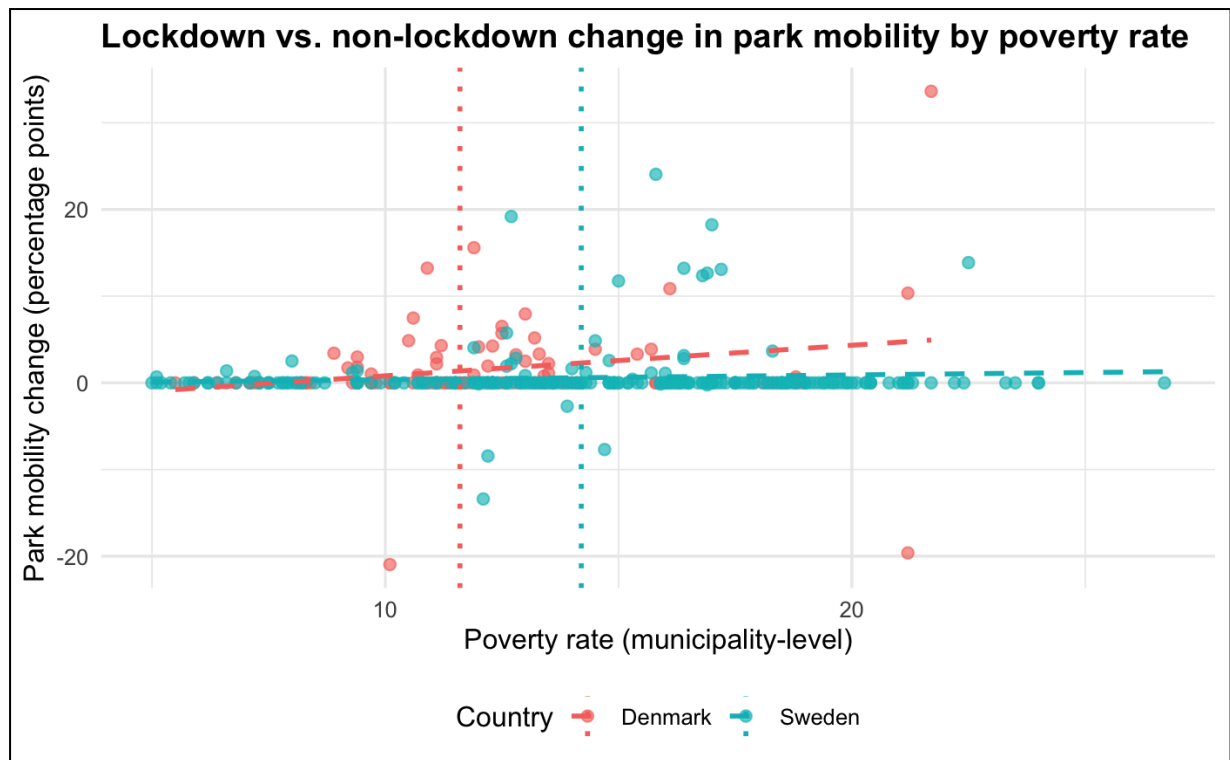
Table 2. Heterogeneous Effects (Median Split)

	Dependent variable	
	Workplace mobility	Park mobility
Intercept	-0.210 [-1.533, 1.112]	-1.184 [-3.010, 0.642]
	-0.312	-1.271
tau: D_it (effect when G_i = 0)	-15.413*** [-16.021, -14.805]	-0.928** [-1.769, -0.087]
	-49.695	-2.162
b1: Treat_i (Denmark)	-4.115*** [-4.449, -3.781]	-0.308 [-0.769, 0.153]
	-24.139	-1.310
b2: High employment (G_i)	0.188 [-0.060, 0.436]	
	1.486	
delta: D_it × High employment	-2.676*** [-3.347, -2.006]	
	-7.821	
b2: High poverty (G_i)		1.919*** [1.577, 2.261]
		10.999
delta: D_it × High poverty		2.528*** [1.602, 3.454]
		5.351
Number of observations	25290	25290
R ²	0.806	0.056
Adj. R ²	0.806	0.053

Note:

Cluster-robust standard errors (municipality level) are in parentheses. Two-sided p-values are shown next to SE and computed using a t distribution with df = (# clusters - 1). 95% confidence intervals are in brackets. Date fixed effects are included but not reported. * p<0.10, ** p<0.05, *** p<0.01.





6. Discussion

This study leverages the policy divergence between Denmark (mandatory lockdown) and Sweden (voluntary recommendations) as a natural experiment to investigate the causal effect of policy strictness on workplace and park mobility. Furthermore, it analyzes how regional socio-economic characteristics, specifically employment and poverty rates, moderate these effects.

6.1 The Differential Effectiveness of Mandates vs. Recommendations (Workplace Mobility)

The results provide strong empirical support for H1. After controlling for date fixed effects, the data indicate that Denmark's mandatory lockdown resulted in a significantly larger decline in workplace mobility—approximately 16.8% more than in Sweden. As illustrated in Figure 1, the sharp decline in mobility in Denmark stands in stark contrast to the flatter curve observed in Sweden. This aligns with the findings of Zhang et al. (2021) and Mishra et al. (2021), suggesting that legally binding interventions are decisively more effective than voluntary behavioral recommendations in reducing economic activity. This implies that when a

government's objective is to rapidly sever virus transmission chains, relying solely on civic responsibility (the Swedish model) is insufficient to curtail workplace commuting; mandatory legal regulations are a powerful tool for altering collective economic behavior.

6.2 Parks as a “Grey Zone” of Universality (Park Mobility)

Regarding park mobility, the results do not support H2, as there was no statistically significant additional increase in Denmark compared to Sweden. However, this does not imply that people avoided parks. Instead, it reveals a more significant phenomenon: as shown in Figure 2, citizens in both countries increased their use of parks following the outbreak. This indicates that regardless of whether policies are strict or lenient, parks serve as a cross-border “grey zone.” When indoor entertainment and socialization are restricted, the demand for outdoor green spaces acts as a universal psychological relief mechanism. Outdoor green spaces become an essential buffer for citizens coping with the stress of lockdowns and maintaining physical and mental well-being.

6.3 Regional Heterogeneity: Economic Hubs and Vulnerable Communities

This study reveals that the policy shock is not uniformly distributed across different regions.

First, the analysis of employment rates supports H1a. In high-employment regions (typically economic hubs), the shock from the mandatory lockdown was more severe, resulting in an additional decline in mobility of approximately 2.7%. This confirms that the lockdown measures disproportionately impacted active commercial areas.

Second, the analysis of poverty rates yielded a counterintuitive but highly insightful finding, leading to the rejection of H2a. While the hypothesis predicted that poverty would limit park access, the data in Table 2 show that high-poverty regions actually exhibited a significant net increase in park mobility relative to low-poverty regions during the lockdown (positive interaction coefficient). This result challenges the traditional view that socio-economic disadvantage necessarily leads to restricted access to green spaces. This finding points to a crucial “Green Equity Paradox.” While theoretical literature suggests lower access to green spaces in poorer areas, our

results demonstrate a higher dependence on them during crises. The “substitution effect” is evident: unlike wealthier households with private gardens or balconies, low-income residents often lack access to private outdoor amenities. Consequently, public parks cease to be a mere leisure option and become an essential “alternative living room”. This inelastic demand highlights that restricted access to public space disproportionately harms vulnerable populations who have nowhere else to go.

Compared to wealthier groups with private outdoor spaces, vulnerable communities have a more urgent and inelastic demand for public green spaces. This highlights the irreplaceable importance of public parks for vulnerable populations during a crisis.

6.4 Limitations

This study is limited by the nature of the Google Community Mobility Reports data, which is derived solely from smartphone users who have enabled Location History. This may not fully represent the behavior of the elderly or extremely low-income groups. Additionally, while Sweden serves as a valid control group, subtle differences in population density and culture between the two nations may still act as confounding variables.

7. Conclusion

7.1. Summary of Findings

By comparing Denmark and Sweden, this study confirms that the impact of COVID-19 policy strictness on human behavior exhibits significant domain-specific differences and regional heterogeneity.

Policy Efficacy: Mandatory lockdowns are decisive in suppressing workplace mobility, significantly outperforming voluntary recommendations.

Human Needs: The demand for parks and outdoor spaces is universal across different policy regimes, indicating that access to nature is a vital psychological coping mechanism during times of crisis.

Social Equity and Unexpected Findings: The Impact of the Policy Shock Is Unequal. Economic centers bear the brunt of business stoppages, while high-poverty communities demonstrate a heightened reliance on public green spaces, rather than restricted use.

7.2. Policy Implications

Based on these findings, future public health crisis management should consider the following key implications:

First, Tiered Intervention and Economic Support: If the goal is to rapidly break transmission chains, legal mandates are more effective than voluntary persuasion. However, policymakers must foresee the substantial shock this will cause to high-employment economic hubs and must simultaneously prepare targeted economic subsidy measures.

Second, Green Justice and Public Health Infrastructure: The finding that high-poverty communities significantly increased their reliance on parks during the lockdown has profound policy implications. Urban planners should regard urban green spaces as critical “public health infrastructure,” equal in importance to medical facilities. Governments should prioritize allocating resources to maintain and expand parks near low-income communities, ensuring these high-density areas have access to safe, high-quality outdoor spaces during future pandemics. This is not merely a leisure issue, but a critical component of building social resilience and mitigating the additional health risks associated with social inequality.

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